

Highlights

Stochastic Physical-Ensemble Quantiles Improve Hybrid Machine-Learning Simulation of Daily Streamflow in the High-Andean Ramis Basin

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- Ensemble quantiles were the dominant driver of hybrid streamflow skill.
- The quantile hybrid improved NSE by 0.311 over stochastic RAMIS.
- Lévy perturbations informed ML features but underdispersed intervals.
- Backend and leakage audits constrained claims for GRU-labelled runs.
- The ablation separated memory, stochasticity, ML and descriptors.

Stochastic Physical-Ensemble Quantiles Improve Hybrid Machine-Learning Simulation of Daily Streamflow in the High-Andean Ramis Basin

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ARTICLE INFO

Keywords:

Daily streamflow simulation
Stochastic hydrological modelling
Physical-machine learning hybridization
Ensemble quantiles
Uncertainty diagnostics
High-Andean basin

ABSTRACT

Hybrid hydrological modelling combines process-based structure with the predictive flexibility of machine learning (ML), but often leaves unclear which component drives improved streamflow prediction. We present StoHyMoLAP, a reproducible and audit-controlled framework for daily streamflow simulation in the high-Andean Ramis basin, within the Lake Titicaca system of southern Peru. The framework couples a lumped RAMIS rainfall-runoff core, an optional linear baseflow reservoir, Lévy-type (α -stable) stochastic perturbations, Monte Carlo physical ensembles, and ML emulators trained from observed forcings or from stochastic-ensemble descriptors. Nine experiments (E0–E8) were evaluated over a common validation window from 8 January 2009 to 31 December 2020 ($n = 4367$ daily samples), with temporal-leakage and backend audits embedded in the evidence chain.

Quantile descriptors from the stochastic physical ensemble provided the largest robust gain. The gradient-boosting quantile hybrid reached NSE = 0.740 and KGE = 0.810, improving NSE by 0.311 relative to the complete stochastic physical model and by 0.165 relative to the pure ML baseline. Replacing ensemble-mean features with ensemble quantiles alone added 0.114 NSE, showing that distributional physical descriptors contain information not captured by a single mean trajectory. The baseflow reservoir modestly improved the deterministic physical baseline, whereas direct Lévy perturbations produced inconsistent point-prediction gains. The stochastic physical bands were strongly underdispersive (PICP = 0.004 against a nominal 0.95 target), so the ensemble is more useful as ML features than as calibrated predictive uncertainty. The numerically best run (NSE = 0.849, KGE = 0.899, RMSE = $0.210 \text{ m}^3 \text{ s}^{-1}$) was labelled as GRU but executed as a fallback multilayer perceptron on flattened sequences; it is not evidence of recurrent-network superiority. These findings support stochastic physical-ML hybridization while showing that uncertainty calibration, low-flow correction and backend-aware reporting remain essential for defensible hydrological modelling.

1. Introduction

Reliable daily streamflow simulation remains a central problem in surface-water hydrology, particularly in basins where observations are sparse, hydroclimatic variability is strong and operational decisions must still be supported by defensible discharge estimates (??). These conditions are characteristic of many high-elevation Andean catchments. In the Peruvian Altiplano, precipitation is strongly seasonal, monitoring networks are sparse relative to the spatial heterogeneity of rainfall, and discharge dynamics are shaped by the alternation between sharp wet-season runoff response and sustained dry-season low flows (???). Such settings expose a persistent limitation of classical rainfall-runoff modelling: even when a conceptual model captures the main water-balance controls, simplified storage structures, uncertain parameters and imperfect meteorological inputs can lead to structural errors that become particularly visible under low-flow, flashy or climatically contrasting conditions (???).

Machine-learning (ML) models offer a complementary route because they can learn nonlinear rainfall-runoff relations directly from data. Large sample studies have shown that recurrent and other data-driven models can match or outperform lumped conceptual models in daily streamflow prediction, including in regional and basin-scale benchmarks (???). However, predictive skill alone does not eliminate hydrological risk. Purely data-driven models may be sensitive to distributional shifts, training-period representativeness, extrapolation to extremes and the absence of

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explicit physical constraints (??). In data-scarce basins, their performance also depends strongly on whether antecedent hydrological state information is available, because rainfall inputs alone may be insufficient to represent catchment memory (??). The practical question is therefore not whether process-based or ML models are universally superior, but how physical structure and empirical flexibility can be combined without obscuring the source of any improvement.

Hybrid physical–ML modelling addresses this question by using process simulations, physically derived states or synthetic physically consistent samples as inputs to an empirical emulator. This strategy has improved runoff simulation in several configurations, including conceptual-model correction, nested hybrid rainfall–runoff modelling, physics-guided deep learning and process-driven deep learning (???????). Most of these studies, however, pass forward a deterministic physical trajectory. That design preserves part of the process information but collapses the conditional distribution of the simulated response into a single value. For stochastic rainfall–runoff transformation, this is a strong simplification: uncertainty in precipitation, model structure and hydrological memory propagates nonlinearly into discharge and can alter both peak timing and magnitude (??). It also limits the ability of the hybrid model to exploit distributional descriptors such as ensemble quantiles, which are closely related to hydrological signatures and to probabilistic forecasting diagnostics (??).

A recent stochastic hybridization study by ? showed that mean and quantile descriptors of a Lévy-driven stochastic HyMoLAP discharge ensemble can improve ML-based rainfall–runoff prediction. This result is important because it moves physical–ML hybridization from deterministic state correction toward distribution-aware physical descriptors. Yet it also leaves three issues unresolved. First, the gain from the hybrid pipeline may come from several components—hydrological memory, stochastic perturbations, the ML correction itself or the use of quantile rather than mean ensemble descriptors—and these effects are not identifiable from a single best-model comparison. Second, a stochastic ensemble that improves point prediction is not necessarily a calibrated predictive interval; coverage, sharpness and interval scores must be evaluated separately. Third, reproducibility claims require that reported model labels match the backend that actually produced the results, especially when deep-learning architectures have software fallbacks or implementation contingencies.

This study addresses these gaps through StoHyMoLAP, an audited stochastic physical–ML framework for daily streamflow simulation in the high-Andean Ramis basin, the principal tributary of Lake Titicaca. The framework couples a lumped RAMIS rainfall–runoff core, an optional linear baseflow reservoir, Lévy-type stochastic perturbations, Monte Carlo physical ensembles and ML emulators trained either from observed forcings or from physical-ensemble descriptors. The purpose is not to introduce another best-performing hybrid model, but to decompose which components are responsible for performance gains. Nine controlled experiments (E0–E8; Table 1) are therefore evaluated over a common validation window using a single workflow (Fig. 1). The ablation design isolates the incremental roles of baseflow memory, stochastic forcing, ML correction and descriptor choice.

The manuscript makes four specific contributions. First, it transfers stochastic physical–ML quantile hybridization to a high-Andean, strongly seasonal and data-scarce Altiplano basin, providing a contrast to previous applications outside this hydroclimatic setting. Second, it shows whether ensemble mean or ensemble quantiles carry the more useful physical information for ML correction. Third, it evaluates the same stochastic ensemble both as a source of predictive features and as a candidate uncertainty band through PICP, PINAW and Winkler diagnostics. Fourth, it embeds temporal-leakage and backend audits into the evidence chain, following the broader call for transparent and reproducible hydrological modelling (?). A key audit finding is stated explicitly: the experiments labelled as GRU in the current run were executed with a fallback multilayer perceptron on flattened sequences, not with a true recurrent backend. They are therefore interpreted as GRU-labelled sequential hybrids under fallback execution, not as evidence of recurrent-network superiority (?).

2. Materials and methods

2.1. Study area

The Ramis River basin is located in the high-Andean Altiplano of southern Peru and drains towards Lake Titicaca. The basin is therefore part of a closed lacustrine system in which river inflow, precipitation and evaporation exert strong control on lake-level variability (???). Previous regional studies show that the hydrology of the Titicaca system is controlled mainly by rainfall and evapotranspiration variability, whereas snowmelt, ice melt and net irrigation withdrawals have a secondary role at the lake-basin scale (?). This regional behaviour makes the Ramis basin a relevant test case for a rainfall–runoff framework designed to combine process-based memory, stochastic forcing and machine-learning correction.

The basin is characterized by high elevation, sparse hydroclimatic monitoring and a strongly seasonal precipitation regime. Rainfall is concentrated during the austral summer wet season, while the dry season is associated with prolonged recession and low-flow persistence (???). Such hydroclimatic asymmetry is challenging for lumped conceptual models because a single storage response may be unable to represent both wet-season peaks and dry-season baseflow. It is also challenging for purely data-driven models, which may learn strong seasonal correlations without representing the storage mechanisms that maintain discharge between rainfall events (??). Daily discharge observations used in this study correspond to the Puente Ramis gauging station operated by SENAMHI-Peru.

2.2. Hydro-meteorological data and temporal protocol

The modelling unit is a daily hydro-meteorological record composed of precipitation (P_t), minimum and maximum temperature ($T_{\min,t}$ and $T_{\max,t}$), potential evapotranspiration (PET_t) and observed streamflow (Q_t^{obs}). The rainfall–runoff core uses an effective precipitation input defined as

$$P_t^{eff} = \max(P_t - PET_t, 0), \quad (1)$$

which provides a non-negative approximation of the water available for runoff generation after atmospheric demand has been accounted for.

All experiments follow the same chronological split. Model calibration and ML training use data from 8 September 1991 to 31 December 2008, whereas validation uses data from 8 January 2009 to 31 December 2020. The final leaderboard is computed over the common validation intersection of all experiments, comprising $n = 4367$ daily samples. This common-window design prevents unequal evaluation support across physical, ML and hybrid configurations and is especially important for experiments that require lagged predictors or sequence construction.

Temporal leakage was controlled at four levels. First, lagged predictors were constructed only from present or past information. Second, sequence samples were generated independently within the training and validation partitions. Third, scalers and transformations used by ML models were fitted exclusively on training data and then applied to validation data. Fourth, flow-regime thresholds used for diagnostic stratification were estimated from training observations only. These controls follow the broader hydrological ML recommendation that model selection and preprocessing must preserve temporal ordering (??). The corresponding audit status is reported in Table 7.

2.3. Lumped rainfall–runoff core

Although the implementation workflow occasionally uses the term “hydrodynamic”, the model evaluated here is a lumped stochastic rainfall–runoff simulator, not a spatially distributed hydraulic or hydrodynamic solver. The process-based component is therefore described as a catchment-scale discharge-dynamics model. Its fast-flow state is represented by the recurrence

$$x_t = \left(1 - \frac{\mu}{\lambda}\right) x_{t-1} + P_t^{eff}, \quad (2)$$

where $\mu > 0$ and $\lambda > 0$ regulate catchment memory and recession stability. The parameter constraint $0 < \mu/\lambda < 1$ ensures a stable, non-oscillatory memory factor.

The fast-flow discharge component is updated as

$$Q_t^{fast} = Q_{t-1}^{fast} - \frac{\mu}{\lambda} Q_0 \left(\frac{\max(Q_{t-1}^{fast}, 0)}{Q_0} \right)^{2\mu-1} + \frac{\alpha_A}{\lambda} x_{t-1} + \sigma Q_{t-1}^{fast} \Delta L_t, \quad (3)$$

where Q_0 is a reference discharge, α_A is an empirical conversion factor, σ controls stochastic amplitude and ΔL_t is a stochastic increment. In deterministic experiments, the stochastic term is inactive. Negative simulated discharge is clipped to zero when non-negative physical post-processing is enabled.

The recurrence in Eqs. 2–3 represents catchment-scale nonlinearity through aggregated storage behaviour. This interpretation is consistent with recession analyses in which nonlinear catchment response can emerge from heterogeneous linear storage elements rather than from a single local nonlinear mechanism (?). In this setting, the parameter μ acts as a shape parameter controlling the effective recession behaviour of the lumped storage system.

2.4. Linear baseflow reservoir

An optional linear reservoir is used to represent hydrological memory beyond the fast-flow recurrence. This addition is motivated by previous work in the Titicaca region showing that groundwater storage fluxes are important for explaining seasonal runoff persistence and dry-season baseflow (?). The reservoir also provides a controlled way to test whether adding a simple memory component improves simulation skill before introducing stochastic perturbations or ML correction.

Recharge to the baseflow store is proportional to effective precipitation:

$$R_t = c_r \alpha_A P_t^{eff}, \quad (4)$$

where $c_r \in [0, 1]$ is the recharge coefficient. Storage and baseflow discharge then evolve as

$$S_{b,t+1} = \max(0, S_{b,t} + R_t - k_b S_{b,t}), \quad (5)$$

$$Q_t^{base} = k_b S_{b,t}, \quad (6)$$

where $k_b \in [0.001, 0.5]$ is the linear recession coefficient. The total process-based discharge is

$$Q_t^{phys} = \max(Q_t^{fast} + Q_t^{base}, 0). \quad (7)$$

The baseflow component is activated in E1, E3 and all hybrid configurations. Its isolated contribution is evaluated through the E1–E0 and E3–E2 ablation contrasts.

2.5. Lévy perturbations and Monte Carlo physical ensembles

Stochastic physical experiments inject α -stable Lévy-type increments into Eq. 3. The perturbation is governed by stability, skewness, location and scale parameters. Heavy-tailed increments are used to represent episodic deviations from the smooth lumped response, which is relevant in a highly seasonal basin where rainfall inputs and streamflow errors are not expected to be Gaussian. This design is consistent with hydrological ensemble studies showing that uncertainty is not limited to the mean response but also involves dispersion, asymmetry and tail behaviour (???).

For each calibrated stochastic configuration, a Monte Carlo ensemble of N trajectories is generated. At each day, the ensemble is summarized by central-tendency, dispersion and quantile descriptors. The quantile set used in the hybrid experiments is

$$\{q_{0.01}, q_{0.05}, q_{0.25}, q_{0.50}, q_{0.75}, q_{0.95}, q_{0.99}\}, \quad (8)$$

together with spread measures such as $q_{0.95} - q_{0.05}$ and $q_{0.75} - q_{0.25}$. Quantile representations are robust summaries of non-Gaussian and fat-tailed distributions, and therefore provide a natural way to pass distributional information from a stochastic physical ensemble to an ML emulator (?). In this study, the same ensemble is evaluated in two roles: as a direct stochastic physical predictor in E2 and E3, and as a source of physically informed ML features in E5–E8.

2.6. Machine-learning emulators and feature sets

The ML component maps a feature vector $\mathbf{z}_t$ to daily simulated discharge,

$$\hat{Q}_t = f_\theta(\mathbf{z}_t), \quad (9)$$

followed by non-negative clipping,

$$Q_t^{sim} = \max(\hat{Q}_t, 0). \quad (10)$$

This post-processing step ensures physically admissible discharge predictions while preserving the fitted ML response over positive flows.

Three feature families are compared. The pure ML baseline (E4) uses observed hydro-meteorological forcings and non-negative lags, including lagged runoff information where allowed by the temporal protocol. Including antecedent runoff is hydrologically meaningful because it acts as a proxy for catchment storage and has been shown to improve rainfall–runoff ML models relative to meteorological predictors alone (?). Mean-based hybrids (E5 and E7) use the mean trajectory of the stochastic physical ensemble. Quantile-based hybrids (E6 and E8) use the ensemble quantiles

Table 1

Controlled experimental matrix. Backend entries report the actual backend used in the current evidence package.

Experiment	Family	Lévy	Baseflow	Emulator	Features	Backend used
E0_RAMIS_DET_NOBF	physical RAMIS	no	no	none	physical only	physical
E1_RAMIS_DET_BF	physical RAMIS	no	yes	none	physical only	physical
E2_RAMIS_LEVY_NOBF	RAMIS–Lévy	yes	no	none	physical only	physical
E3_RAMIS_LEVY_BF	RAMIS–Lévy	yes	yes	none	physical only	physical
E4_ML_PURE_XGB	pure ML	no	no	XGB-labelled	observed forcings	sklearn_hgb
E5_HYB_XGB_MEAN	hybrid	yes	yes	XGB-labelled	ensemble mean	sklearn_hgb
E6_HYB_XGB_QUANTILES	hybrid	yes	yes	XGB-labelled	ensemble quantiles	sklearn_hgb
E7_HYB_GRU_MEAN	sequential hybrid	yes	yes	GRU-labelled	mean sequence	fallback MLP
E8_HYB_GRU_QUANTILES	sequential hybrid	yes	yes	GRU-labelled	quantile sequence	fallback MLP

and spread descriptors listed in Eq. 8. This design tests whether the ML emulator benefits primarily from a corrected central physical trajectory or from richer distributional descriptors of the physical ensemble.

The tabular emulator is labelled as XGB in the experiment names. In the current evidence package, however, the actual backend was `HistGradientBoostingRegressor` from `scikit-learn`, reported in the audit tables as `sklearn_hgb`. The corresponding results are therefore interpreted as gradient-boosted tree hybrids rather than as package-specific evidence for `xgboost`. Hyperparameter search used temporally ordered cross-validation, avoiding random shuffling of time steps during model selection.

The sequential emulator is labelled as GRU in the experiment names and is designed to use recurrent layers when a supported backend is available. In the present execution, the recurrent backend was not available; the implementation flattened each input sequence and trained a multilayer perceptron instead. Consequently, E7 and E8 are retained in the experimental matrix because they evaluate sequentially constructed features, but they are interpreted throughout the manuscript as GRU-labelled fallback-MLP runs, not as evidence of true recurrent GRU superiority (??).

2.7. Calibration objective

The process-based and stochastic configurations are calibrated with a scalar objective that combines efficiency, distributional agreement, bias and interval coverage:

$$J = \frac{w_{NSE}(1 - NSE) + w_{KGE}(1 - KGE) + w_{PBIAS} \left| \frac{PBIAS}{100} \right| + w_{cov}|c^* - PICP|}{w_{active}}, \quad (11)$$

where $c^* = 0.95$ is the nominal coverage target and w_{active} is the sum of the active weights. The default weights are $w_{NSE} = 0.40$, $w_{KGE} = 0.30$, $w_{PBIAS} = 0.15$ and $w_{cov} = 0.15$. When a model does not produce prediction intervals, the coverage term is omitted and the remaining weights are renormalized.

The composite objective is used because no single deterministic metric fully describes hydrological model performance, particularly when both high-flow peaks and low-flow persistence are relevant (?). The coverage term is included to avoid selecting stochastic configurations that are sharp but severely underdispersive, a common limitation in hydrological ensemble prediction (??).

2.8. Experimental matrix and ablation design

The experimental matrix was designed to isolate the contribution of each modelling component under a common validation window. E0 and E1 compare the deterministic RAMIS core without and with the baseflow reservoir. E2 and E3 add Lévy stochasticity without and with baseflow. E4 is a pure ML baseline trained only from observed predictors. E5 and E6 are gradient-boosting hybrids using, respectively, ensemble-mean and ensemble-quantile physical descriptors. E7 and E8 use the same mean and quantile feature families in a sequential-feature setting, but with the fallback MLP backend described above.

This matrix supports five planned contrasts: the effect of baseflow (E1–E0 and E3–E2), the direct effect of stochastic perturbations under baseflow (E3–E1), the effect of pure ML correction relative to the full stochastic physical model (E4–E3), the effect of hybridization (E5–E3 and E6–E3), and the value of quantile descriptors relative to mean descriptors (E6–E5 and E8–E7). The same validation samples are used in every contrast.

2.9. Relation to prior stochastic physical–ML hybridization

The idea of feeding descriptors from a stochastic physical ensemble into an ML emulator is closely related to the hybridization strategy of ?. The present study is framed as an ablation and audit extension of that idea. Rather than reporting only the best hybrid configuration, it decomposes the pipeline into physical memory, stochastic perturbation, ML correction and descriptor choice. It also evaluates whether the same ensemble that improves ML features can serve as a calibrated predictive uncertainty band. This distinction is important because physically plausible ensemble spread does not necessarily imply reliable probabilistic coverage (??).

2.10. Evaluation metrics and audit protocol

Point-prediction performance is evaluated with Nash–Sutcliffe efficiency (NSE), Kling–Gupta efficiency (KGE), root mean squared error (RMSE), mean absolute error (MAE), percent bias (PBIAS) and the coefficient of determination (R^2). Regime-specific metrics are computed for low, mid and high flows using thresholds estimated from training observations only. Low-flow NSE is interpreted cautiously because efficiency scores based on squared error are sensitive to low observed variance and can penalize dry-season errors in ways that are not proportional to their hydrological relevance (?).

Stochastic reference bands are evaluated using prediction interval coverage probability (PICP), prediction interval normalized average width (PINAW), mean prediction interval width (MPIW) and Winkler score. The intended behaviour is high coverage with the narrowest possible intervals, but sharpness is not interpreted as skill when coverage is far below the nominal level (???). The final audit checks chronological split integrity, feature construction, target leakage, backend metadata, common-window consistency and non-negative post-processing. The audit results are reported in Table 7.

3. Results

3.1. Common-window ranking

All experiments were evaluated on the same validation window defined in Section 2.2. Table 2 and Fig. 2 show a clear separation between the physical-only baselines, the pure ML baseline and the hybrid configurations. The deterministic and stochastic physical baselines (E0–E3) clustered within a narrow NSE range of 0.414–0.438, whereas the pure forcing-based ML model (E4) reached $NSE = 0.575$. All hybrid models exceeded E4 in NSE, indicating that the stochastic physical ensemble supplied additional predictive information beyond the observed forcing variables alone.

The strongest *backend-consistent* configuration was E6_HYB_XGB_QUANTILES, which achieved $NSE = 0.740$, $KGE = 0.810$, $RMSE = 0.275 \text{ m}^3 \text{ s}^{-1}$, $MAE = 0.162 \text{ m}^3 \text{ s}^{-1}$ and $PBIAS = 0.927\%$. This run is the main evidence for the proposed quantile-based stochastic physical–ML hybridization because its manuscript label and executed backend are consistent. The numerically best run was E8_HYB_GRU_QUANTILES ($NSE = 0.849$, $KGE = 0.899$ and $RMSE = 0.210 \text{ m}^3 \text{ s}^{-1}$), but it is interpreted separately because the audit showed that it ran as a fallback multilayer perceptron on flattened sequences rather than as a recurrent GRU. The ranking is therefore reported in two layers: E8 is the best numerical configuration, whereas E6 is the strongest robust configuration for defensible process-based interpretation.

3.2. Hydrograph and flow-duration behaviour

The validation hydrograph in Fig. 3 shows that the hybrid models reproduce the timing and amplitude of seasonal streamflow fluctuations more closely than the physical-only baselines. The physical models retain the main wet-season response but frequently collapse towards near-zero discharge during recession periods. This behaviour is also evident in the flow-duration curves (Fig. 4), where E0 shows an unrealistic decline in the high-exceedance tail. Adding the baseflow reservoir reduces this problem only partially, indicating that the single linear storage term improves recession persistence but does not fully resolve dry-season memory.

The hybrid configurations provide a closer match to the observed flow-duration curve over most exceedance probabilities. The largest visual improvement occurs in the lower-flow tail, where quantile-based hybrids avoid the extreme collapse of the deterministic physical trajectory. Nevertheless, the hydrograph also shows remaining errors during sharp wet-season peaks. Thus, the hybrid improvement is not a complete correction of event-scale dynamics; it is mainly a reduction of systematic distributional errors and dry-season underestimation.

Table 2

Final common-window ranking. Metrics are computed on 4367 validation samples from 8 January 2009 to 31 December 2020.

Rank	Experiment	NSE	KGE	RMSE	MAE	PBIAS	Backend
1	E8_HYB_GRU_QUANTILES	0.849	0.899	0.210	0.124	1.579	fallback_mlp_on_flattened_sequences
2	E6_HYB_XGB_QUANTILES	0.740	0.810	0.275	0.162	0.927	sklearn_hgb
3	E7_HYB_GRU_MEAN	0.686	0.789	0.302	0.192	1.644	fallback_mlp_on_flattened_sequences
4	E5_HYB_XGB_MEAN	0.626	0.715	0.330	0.210	2.007	sklearn_hgb
5	E4_ML_PURE_XGB	0.575	0.661	0.352	0.223	0.742	sklearn_hgb
6	E1_RAMIS_DET_BF	0.438	0.714	0.405	0.248	0.860	physical
7	E3_RAMIS_LEVY_BF	0.429	0.725	0.408	0.255	-1.218	physical
8	E2_RAMIS_LEVY_NOBF	0.427	0.717	0.409	0.268	-7.551	physical
9	E0_RAMIS_DET_NOBF	0.414	0.713	0.413	0.270	-5.865	physical

Table 3

Ablation effects on the common validation window. Positive Δ NSE and Δ KGE indicate improvement; negative Δ RMSE and Δ MAE indicate error reduction.

Ablation	Baseline	Candidate	Δ NSE	Δ KGE	Δ RMSE
deterministic_baseflow	E0_RAMIS_DET_NOBF	E1_RAMIS_DET_BF	0.024	0.000	-0.009
stochastic_baseflow	E2_RAMIS_LEVY_NOBF	E3_RAMIS_LEVY_BF	0.002	0.008	-0.001
levy_given_baseflow	E1_RAMIS_DET_BF	E3_RAMIS_LEVY_BF	-0.010	0.011	0.004
pure_ml_vs_physical	E3_RAMIS_LEVY_BF	E4_ML_PURE_XGB	0.147	-0.064	-0.056
hybrid_mean_vs_physical	E3_RAMIS_LEVY_BF	E5_HYB_XGB_MEAN	0.198	-0.010	-0.078
hybrid_quantiles_vs_physical	E3_RAMIS_LEVY_BF	E6_HYB_XGB_QUANTILES	0.311	0.085	-0.133
quantiles_vs_mean_xgb	E5_HYB_XGB_MEAN	E6_HYB_XGB_QUANTILES	0.114	0.095	-0.055
gru_mean_vs_xgb_mean	E5_HYB_XGB_MEAN	E7_HYB_GRU_MEAN	0.060	0.074	-0.028
gru_quantiles_vs_xgb_quantiles	E6_HYB_XGB_QUANTILES	E8_HYB_GRU_QUANTILES	0.109	0.089	-0.066

3.3. Ablation effects

Table 3 and Fig. 5 isolate the contribution of each modelling component. Adding the baseflow reservoir to the deterministic physical model increased NSE by 0.024 and reduced RMSE by $0.009 \text{ m}^3 \text{ s}^{-1}$ (E1–E0). Under stochastic forcing, the same structural addition produced a smaller but still positive gain in NSE and KGE (E3–E2). In contrast, adding Lévy-type perturbations to the baseflow-enabled physical model did not improve point-prediction skill: NSE decreased by 0.010 and RMSE increased by $0.004 \text{ m}^3 \text{ s}^{-1}$, although KGE increased slightly. This confirms that stochastic perturbation is not, by itself, the main source of point-prediction improvement.

The dominant gain came from hybridization. Relative to the complete stochastic physical baseline (E3), the pure forcing-based ML model (E4) improved NSE by 0.147, whereas the XGB-labelled hybrid using ensemble means (E5) improved NSE by 0.198. The quantile-based hybrid (E6) produced the largest backend-consistent ablation effect, with Δ NSE = 0.311 and Δ RMSE = $-0.133 \text{ m}^3 \text{ s}^{-1}$ relative to E3. The direct comparison between E6 and E5 added 0.114 NSE and 0.095 KGE, showing that ensemble quantiles carry useful predictive information that is lost when the stochastic physical ensemble is compressed to its mean alone.

The GRU-labelled contrasts must be interpreted with the backend audit in mind. E7 and E8 improved over their XGB-labelled counterparts, but these gains cannot be attributed to recurrent-network memory because both runs used a fallback MLP backend. They are therefore reported as additional sequential-feature experiments under fallback execution, not as evidence that GRU architecture is superior.

3.4. Performance by flow regime

Regime-specific RMSE values (Table 4 and Fig. 6) show that model skill is strongly flow-dependent. The physical-only configurations had similar errors across the selected regimes, with the largest absolute errors in the high-flow class. The pure ML model reduced mid-flow RMSE relative to the physical baselines, but its low-flow RMSE remained close to that of the physical models. The quantile-based hybrid E6 substantially reduced errors in all regimes, and the numerically best E8 configuration further reduced RMSE under fallback-MLP execution.

Table 4

RMSE ($\text{m}^3 \text{s}^{-1}$) by flow regime for representative models. Regime thresholds were estimated from calibration observations only.

Experiment	Low	Mid	High
E0_RAMIS_DET_NOBF	0.177	0.392	0.615
E1_RAMIS_DET_BF	0.197	0.365	0.613
E3_RAMIS_LEVY_BF	0.185	0.387	0.604
E4_ML_PURE_XGB	0.189	0.276	0.566
E6_HYB_XGB_QUANTILES	0.075	0.222	0.461
E8_HYB_GRU_QUANTILES	0.047	0.171	0.352

Table 5

Uncertainty diagnostics for stochastic physical reference bands.

Experiment	PICP	PINAW	MPIW	Winkler
E2_RAMIS_LEVY_NOBF	0.160	0.084	0.219	7.297
E3_RAMIS_LEVY_BF	0.004	0.003	0.009	10.050

Table 6

Backend-aware manuscript notes for ML and hybrid configurations.

Experiment	ML label	Backend label	Feature source	Manuscript claim
E4_ML_PURE_XGB	xgboost	sklearn_hgb	observed_forcing	report as gradient boosting (HGB)
E6_HYB_XGB_QUANTILES	xgboost	sklearn_hgb	stochastic_ensemble	report as gradient boosting (HGB)
E7_HYB_GRU_MEAN	gru	fallback_mlp_on_flattened_sequences	stochastic_ensemble	fallback MLP, not real recurrent GRU
E8_HYB_GRU_QUANTILES	gru	fallback_mlp_on_flattened_sequences	stochastic_ensemble	fallback MLP, not real recurrent GRU

The largest proportional improvement occurred in the low-flow regime: RMSE decreased from $0.177 \text{ m}^3 \text{s}^{-1}$ for E0 to $0.075 \text{ m}^3 \text{s}^{-1}$ for E6 and $0.047 \text{ m}^3 \text{s}^{-1}$ for E8. The mid-flow and high-flow regimes also improved, but larger residual errors persisted during high flows, where event timing and peak magnitude remain more difficult to reproduce. These results support the interpretation from the flow-duration curves: stochastic ensemble quantiles are most useful for correcting systematic distributional errors, especially during recession and low-flow periods, while peak simulation remains only partially corrected.

3.5. Uncertainty diagnostics and backend audit

The stochastic physical intervals were strongly underdispersive (Table 5 and Fig. 7). E2 achieved PICP = 0.160 and E3 achieved PICP = 0.004, both far below the nominal 0.95 target. The corresponding PINAW values were also small, particularly for E3 (PINAW = 0.003), indicating that the intervals were narrow rather than reliably calibrated. The high Winkler scores confirm that most observed values fell outside the nominal intervals. Consequently, the stochastic ensemble should not be reported as a calibrated predictive uncertainty product in its current form. Its main demonstrated value is as a source of physically structured features for ML correction.

Table 6 summarizes the backend audit. The XGB-labelled experiments (E4–E6) used the `sklearn_hgb` backend, whereas the GRU-labelled experiments (E7 and E8) used `fallback_mlp_on_flattened_sequences`. This distinction changes the interpretation of the leaderboard: E8 may be reported as the best numerical experiment, but not as evidence for recurrent neural networks. The reproducibility checklist in Table 7 returned no FAIL item. WARN items were associated with missing observed-flow values, low-flow physical limitations, optional figure entries and the fallback backend, rather than with direct evidence of temporal leakage.

Table 7

Reproducibility checklist for the current evidence package.

Item	Status	Evidence
Common validation intersection	PASS	leaderboard_common_intersection.csv
Anti-leakage audit	WARN	leakage_audit_status.txt=WARN
Physical audit	WARN	physical_audit_status.txt=WARN
Figure manifest	PASS	outputs/figures/phase34/phase34_figure_manifest.csv
GRU wording	WARN	E7/E8 use fallback backend
Optional physical components in figures	WARN	skipped entries in phase34_figure_manifest.csv

4. Discussion

4.1. Hydrological memory and the role of baseflow

The positive deterministic contrast E1–E0 confirms that adding a linear baseflow reservoir improves the physical RAMIS baseline, in agreement with the modelling evidence from semi-arid Andean catchments reviewed by ? : “the best simulations could be achieved with separate parameter sets for varying hydro-climatic conditions,” and a dedicated slow-drainage component is necessary to represent the recession-dominated dry season. The NSE and RMSE improvements are modest in absolute terms but hydrologically meaningful: the baseflow reservoir introduces the delayed drainage memory that cannot be captured by an instantaneous fast-flow response alone. This conclusion echoes the quantitative demonstration of ? that including runoff memory in model inputs is “appropriate and contributes to an improvement of the models’ accuracy.”

However, the regime results in Table 4 show that low-flow simulation remains challenging despite the presence of the baseflow reservoir. A single linear store can represent average recession behaviour but is insufficient when the storage is heterogeneous across the catchment, as would be expected in an Altiplano landscape with wetlands, peatlands (bofedales) and variable permeability (?). ? showed theoretically that “between-hillslope heterogeneity alone can give rise to apparently nonlinear recession slope curves,” implying that the effective storage–discharge relationship inferred from recession analysis will always appear more complex than a simple linear reservoir. A practical consequence for StoHyMoLAP is that the baseflow improvement is bounded by the structural limitation of a single-reservoir representation. This is consistent with the finding of ? that “parallel reservoirs with independent time constants increase model performance in groundwater-dominated catchments,” suggesting a natural extension for future work.

The limited low-flow improvement also reflects a fundamental measurement challenge: the NSE criterion assigns most weight to high-flow variance (?), so the global NSE improvement from the baseflow reservoir is modest even when the absolute error in the low-flow regime is reduced. Following the recommendation of ? to use the NSE on inverse flows (NSE_{iQ}^*) to focus evaluation on the lowest 20% of flows, future studies should track this metric explicitly in addition to global NSE.

4.2. Lévy perturbations as stochastic descriptors rather than calibrated uncertainty

Lévy perturbations modify the distributional behaviour of physical simulations and, in principle, can represent the heavy-tailed variability documented for Andean precipitation and streamflow. The coverage diagnostics in Table 5, however, demonstrate that the resulting stochastic bands are strongly underdispersive: PICP values of 0.16 and 0.004 are far below the nominal 0.95, and Winkler scores reflect large penalties for missing observations. This failure to achieve calibrated coverage is not unique to Lévy-driven models: ? showed that even sophisticated multi-model ensemble systems “fail to maintain the required dispersion throughout the entire forecast horizon,” and ? found that correcting precipitation inputs alone is “obviously not enough to achieve streamflow forecast reliability.” The more general point is that calibrated uncertainty requires explicitly accounting for all sources of error simultaneously—not only stochastic forcing, but also parameter uncertainty, structural uncertainty and initial-condition uncertainty (??).

The underdispersive nature of the bands does not negate the value of Lévy perturbations within StoHyMoLAP; it merely redirects that value. Rather than serving as calibrated intervals, the stochastic ensemble summaries—particularly the quantile descriptors in Eq. 8—function as informative features that encode distributional information about the conditional response of the physical model. The key insight from ? is directly applicable here: “the spectral slopes of the rain fields play an essential role in determining the hydrological response,” and features that preserve asymmetry and tail behaviour (quantiles) carry more information than features that collapse the distribution to its

mean. ? showed that quantile representations are “less sensitive to fat-tailed distributions and outliers,” a property that is particularly desirable when the ensemble members are driven by α -stable noise.

4.3. Why quantile-based hybridization works

The most robust empirical signal in this study is the advantage of quantile-based hybridization. The E6–E5 contrast adds 0.114 NSE and 0.095 KGE purely from switching the physical descriptor from a mean to a quantile vector, holding everything else constant. This result is interpretable within the framework of ? : “there are two ways for improving predictive performance—improving the prediction algorithm and discovering new informative predictors,” and quantile descriptors of the physical ensemble constitute precisely the latter. A single ensemble mean retains first-moment information but discards the conditional spread, asymmetry and tail behaviour of the physical simulation. In the Ramis basin, where the flow regime alternates sharply between wet-season peaks and dry-season recession, these distributional properties are not constant: the upper quantiles expand during the wet season while the lower quantiles collapse. This asymmetric dynamics is informationally invisible to a mean-only descriptor and yet is directly observable in the quantile vector.

The hybrid emulator (E6) also outperforms the pure ML baseline (E4) despite E4 having access to observed meteorological forcing. This indicates that physically simulated ensemble descriptors provide information beyond what is contained in the observed forcings alone, a result consistent with ? who found NSE improvements of 23% when ML surrogates were enriched with synthetic physical data, and with ? who showed that SWAT outputs “provide a proxy about the instantaneous wetness condition of the catchment.” The interpretation is that the physical ensemble descriptors effectively encode catchment antecedent state in a form that the ML emulator can more readily exploit than raw meteorological time series. This is consistent with the argument of ? that “a machine learning model needs a teacher (hydrologist) that provides it with textbooks and then leaves it for self-teaching,” and with the finding of ? that models lacking antecedent runoff information cannot compete with physically constrained alternatives.

One should note, however, that the E6 advantage over E4 is partly confounded with the difference in feature type (ensemble statistics versus raw forcing), and the E6–E4 contrast alone cannot distinguish between the value of hydrological structure and the value of temporal aggregation. Resolving this would require an additional ablation experiment pairing observed forcing with quantile features derived from simple lag statistics, which is left for future work. This caveat is consistent with the general warning of ? that “implicit coupling... requires more elaborate design of the model structure and inputs, as well as more thorough verification of the results to avoid learning other misleading relationships.”

4.4. Backend-aware interpretation of the sequential experiments

E8 is the top-ranked configuration in Table 2, but the result must be interpreted together with Table 6. The fallback MLP backend in E7 and E8 processes flattened sequences rather than recurrent hidden states, and therefore does not implement the temporal gating mechanism that is the defining feature of GRU cells (??). The valid conclusion is narrower: a GRU-labelled sequential hybrid using quantile descriptors and implemented with a fallback flattened-sequence MLP achieved the best validation metrics in the current environment. Whether a true recurrent GRU would reproduce or exceed this performance cannot be established from the current evidence package.

The consistent ordering of quantile over mean descriptors in both the XGB-labelled (E6>E5) and the GRU-labelled (E8>E7) families suggests that the descriptor choice is the driver of improvement rather than the emulator architecture. This is an important practical conclusion: it implies that the gain from quantile hybridization should be architecture-agnostic, and therefore reproducible under a true GRU backend. Future work should rerun E7 and E8 with TensorFlow or PyTorch backends and compare the GRU-specific gain to the descriptor-specific gain measured here. Transparent backend reporting is essential for such comparison, as emphasized by ? : “open hydrologists document all components of their data collection and analysis pipeline, favoring open and non-proprietary technologies.”

4.5. Leakage, reproducibility and submission readiness

The audit status in Table 7 strengthens the comparison by documenting the major leakage controls: strict temporal splitting, non-negative lags, sequence construction within temporal subsets, train-only scalars and train-only regime thresholds. “Only 1% of hydrology papers were fully reproducible” in a recent large-scale survey (?), and the StoHyMoLAP audit is an explicit step toward closing that gap.

The WARN status should remain visible in the manuscript or supplementary material. WARN does not indicate detected temporal leakage, but it does identify issues that affect interpretation: missing observed-flow values, physical

low-flow limitations, underdispersive stochastic bands and fallback ML backends. These are honest disclosures consistent with the open-science principles advocated by ? and with the rigorous benchmarking framework recommended by ?.

5. Limitations

This study has five main limitations. First, the sequential experiments labelled as GRU were not executed with a real recurrent backend, so their results cannot support claims about recurrent neural networks (?). Second, the Lévy stochastic bands are underdispersive and should not be treated as calibrated uncertainty intervals; recalibration following the guidelines of ? and ? is required before these bands can be used operationally. Third, low-flow bias remains substantial despite improvements in absolute-error metrics; future work should apply the inverse-flow NSE criterion of ? to focus calibration explicitly on the dry-season regime. Fourth, the audit status is WARN rather than fully PASS, requiring transparent reporting of missing observations and physical-audit warnings. Fifth, confirm data provenance and units (Section 2.1): if the current package corresponds to an illustrative or reduced-computation run, or if the discharge series is normalized or refers to a sub-catchment rather than the full Puente Ramis basin, the final submission must regenerate the complete evidence package with the definitive, traceable observed Ramis dataset (source, station, units) and publication-level calibration settings.

6. Conclusions

StoHyMoLAP provides a reproducible, audit-controlled framework for evaluating stochastic physical-machine-learning hybridization in daily streamflow simulation, applied here to the high-Andean Ramis basin in the Lake Titicaca system. Four main conclusions emerge from the controlled ablation.

First, quantile descriptors from the stochastic physical ensemble are the dominant driver of hybrid improvement: replacing the ensemble mean with quantile features added 0.114 NSE (E6–E5), and the full quantile hybrid surpassed the physical model by 0.311 NSE (E6–E3). This confirms, in an Altiplano setting, the argument of ? and the quantitative demonstration of ? that distributional descriptors carry information beyond a single trajectory.

Second, the baseflow reservoir improves the deterministic physical baseline (E1–E0: $\Delta\text{NSE} = 0.024$) by introducing slow-drainage memory consistent with the groundwater-dominated response of the Ramis system (?), but a single linear reservoir is insufficient to fully correct the low-flow bias documented by ? and ?.

Third, Lévy perturbations improve point prediction inconsistently as a direct physical forcing (E3–E1: $\Delta\text{NSE} = -0.010$) but contribute positively as generators of ensemble diversity whose distributional summaries serve as informative ML features. The stochastic bands are strongly underdispersive ($\text{PICP} = 0.004$ for E3) and cannot be used as calibrated predictive intervals without further post-processing (??).

Fourth, the numerically top-ranked configuration (E8, $\text{NSE} = 0.849$) must be reported as a GRU-labelled fallback-MLP run rather than as evidence of recurrent neural-network superiority (?). The descriptor ordering (quantile > mean) is nevertheless consistent across both emulator families, supporting the conclusion that the information value of distributional physical descriptors is architecture-agnostic.

Future work should regenerate E7 and E8 with real recurrent backends, recalibrate stochastic uncertainty toward nominal coverage, apply inverse-flow NSE criteria to improve low-flow performance, and evaluate the framework at full-resolution Ramis data before transferring to neighbouring Altiplano catchments.

Code and data availability

The manuscript tables and figures are generated from the StoHyMoLAP Phase 3.4B–3.5 result package. The scripts `make_method_diagram.py` and `build_paper_assets.py` reproduce all figures and CSV tables. Backend metadata should be regenerated after any change to the ML environment. repository URL/DOI and the SENAMHI/ANA data-source citation.

Declaration of generative AI and AI-assisted technologies in the writing process

During preparation of this draft, AI-assisted writing was used to organize the manuscript, convert reproducible artefacts into narrative text, and generate supporting LaTeX scripts. The authors remain responsible for verifying all scientific claims, numerical values, references, data provenance and final wording before submission.

CRedit authorship contribution statement

Jose Zevallos: Conceptualization of this study, Methodology, Software, Writing – original draft.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

CRedit authorship contribution statement

Jose Zevallos: Conceptualization of this study, Methodology, Software, Writing – original draft.

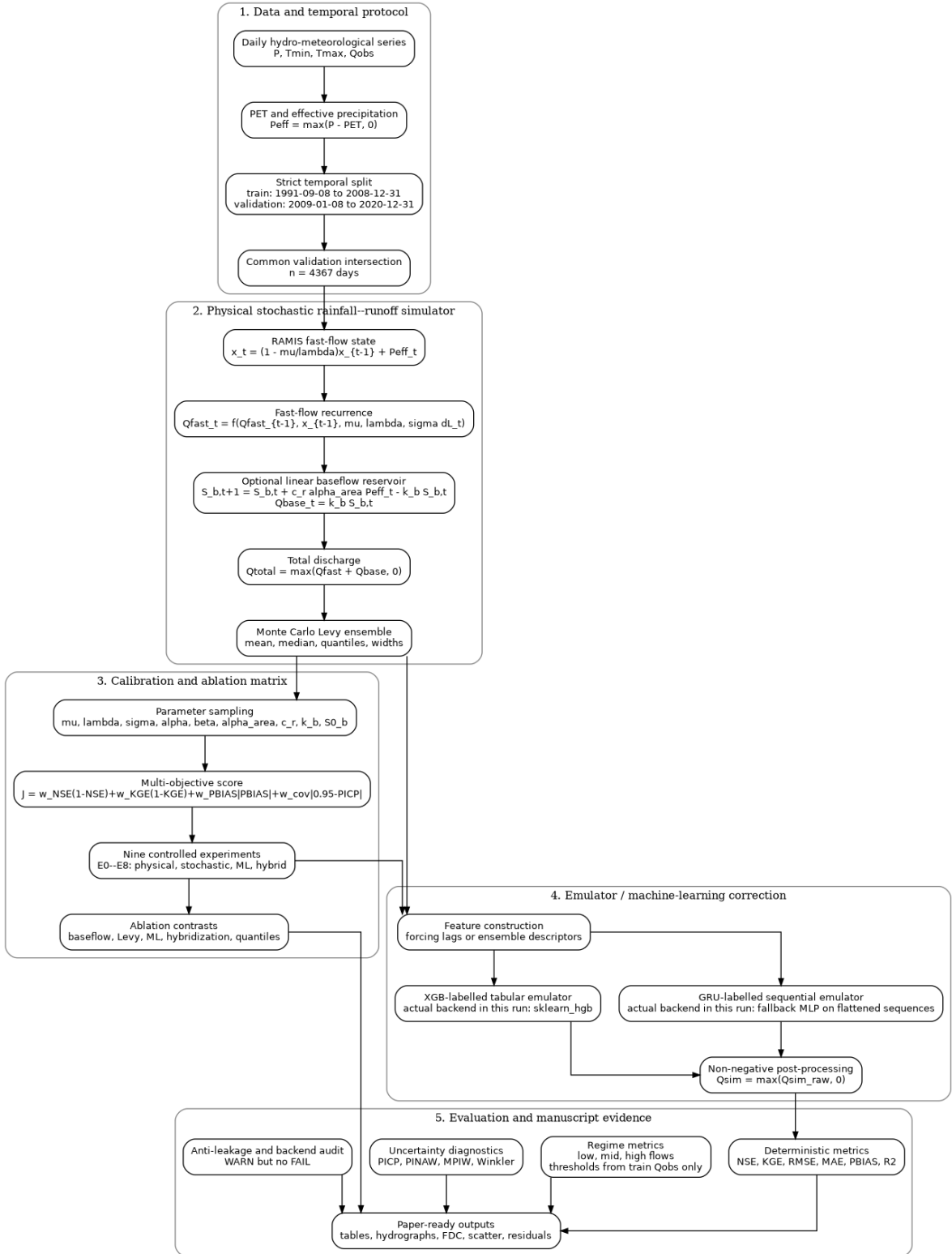


Figure 1: StoHyMoLAP workflow from daily hydro-meteorological data to lumped physical simulation, Lévy-type stochastic ensemble generation, ensemble-descriptor construction, ML emulation, ablation comparison and audit-controlled manuscript evidence.

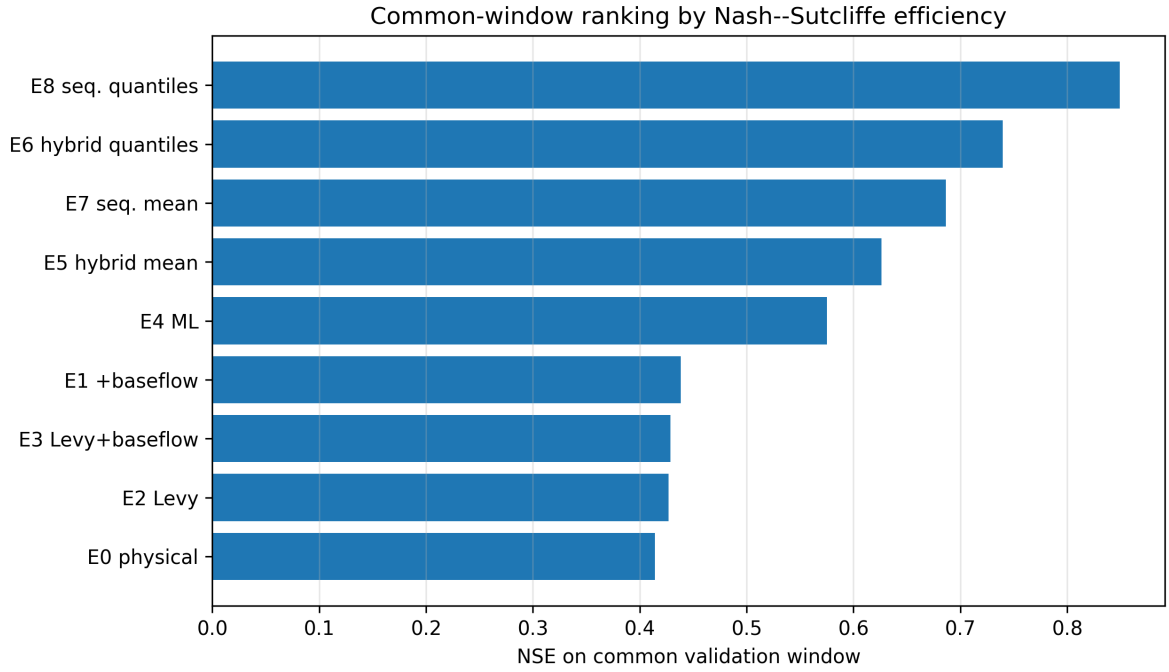


Figure 2: Ranking of all experiments by common-window NSE. Numerical values correspond to Table 2.

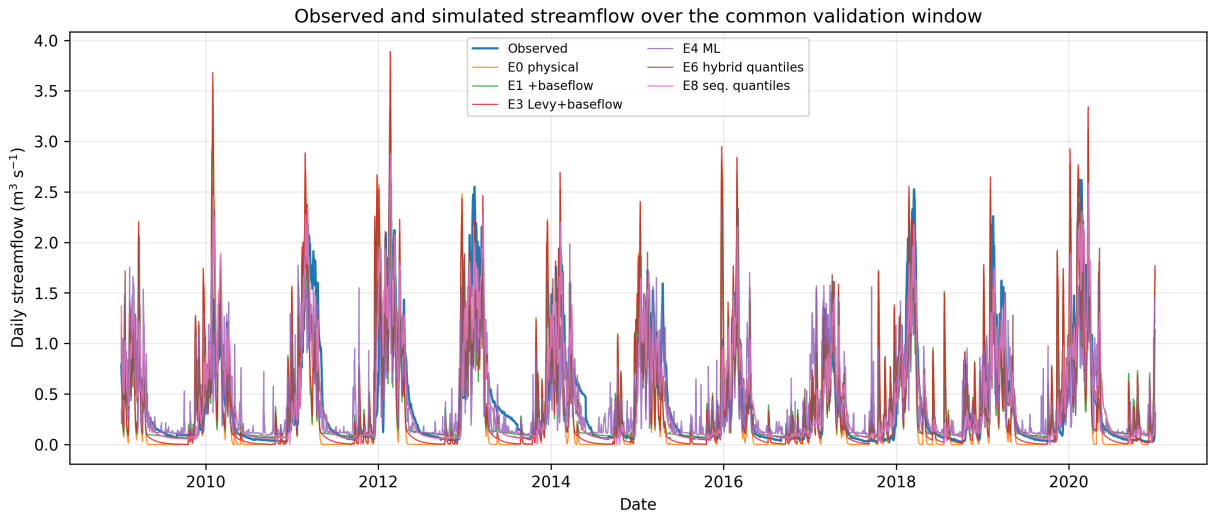


Figure 3: Observed and simulated daily discharge over the common validation window for representative physical, pure ML and hybrid models.

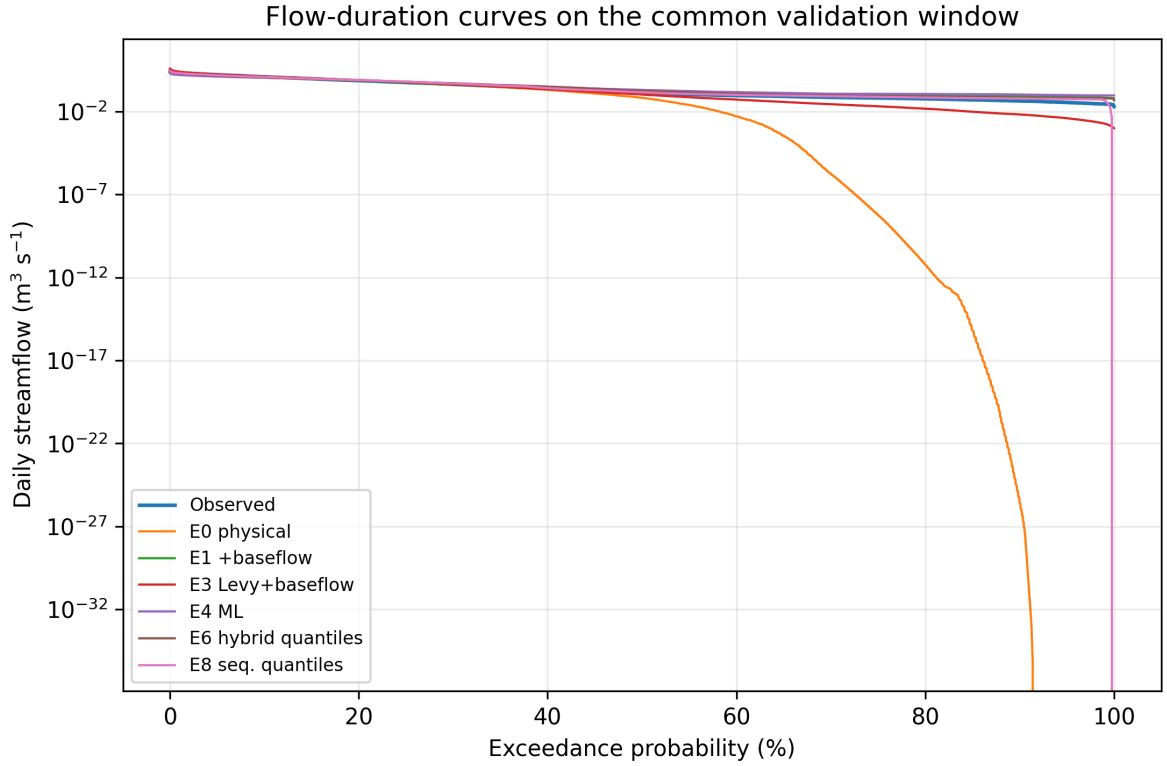


Figure 4: Flow-duration curves for observed discharge and representative simulations over the common validation window.

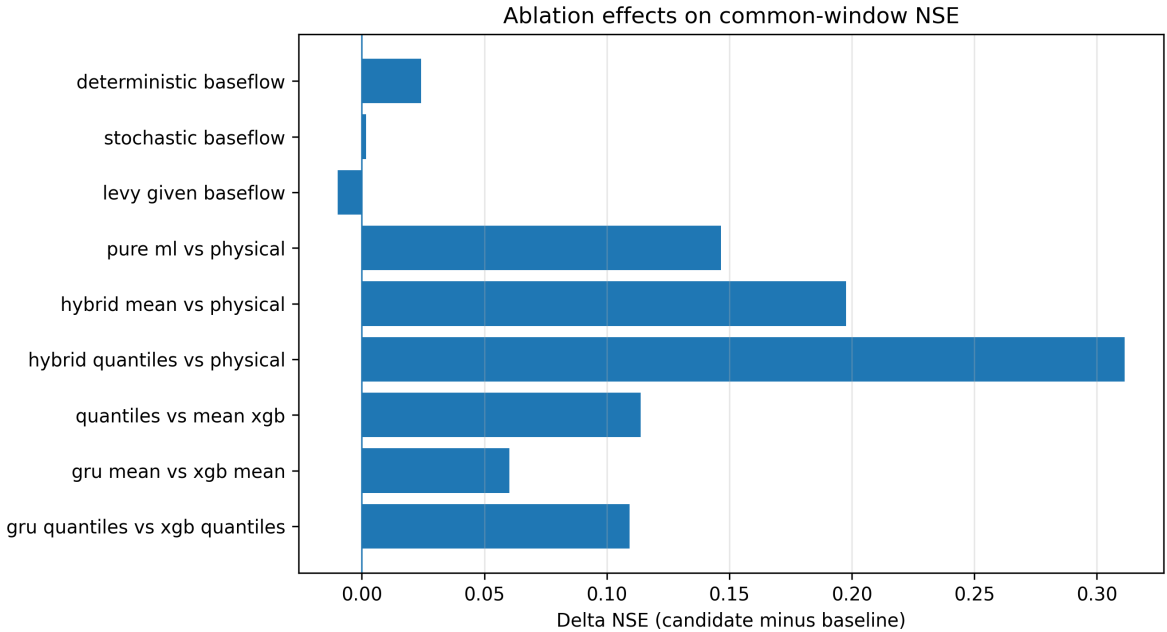


Figure 5: Ablation effects expressed as candidate-minus-baseline ΔNSE . Numerical values are reported in Table 3.

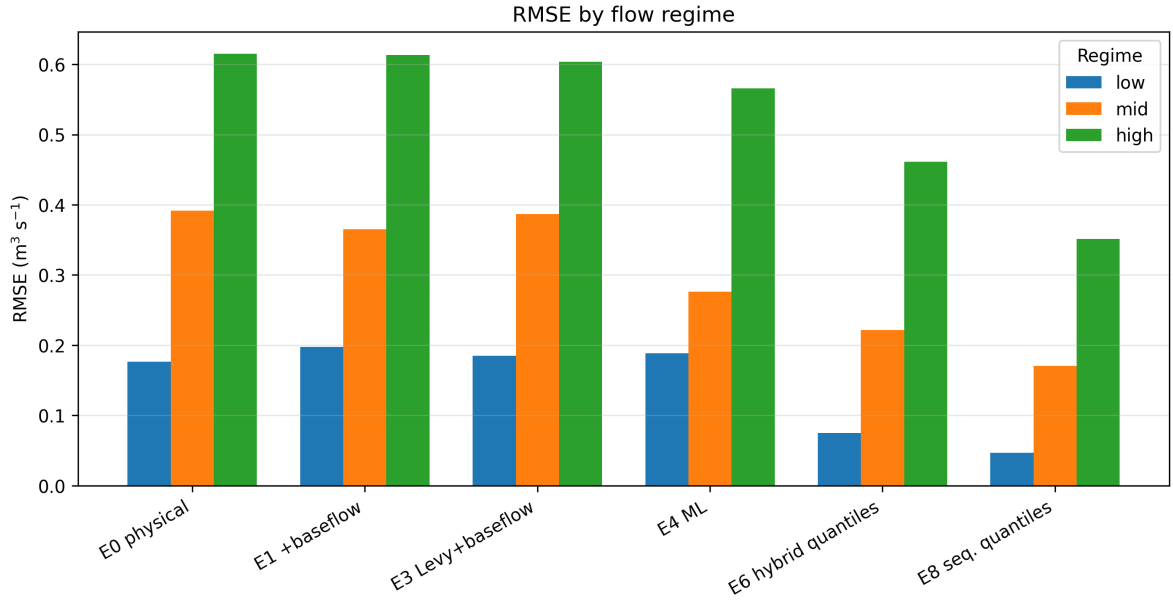


Figure 6: RMSE by low, mid and high flow regimes for representative models. Values correspond to Table 4.

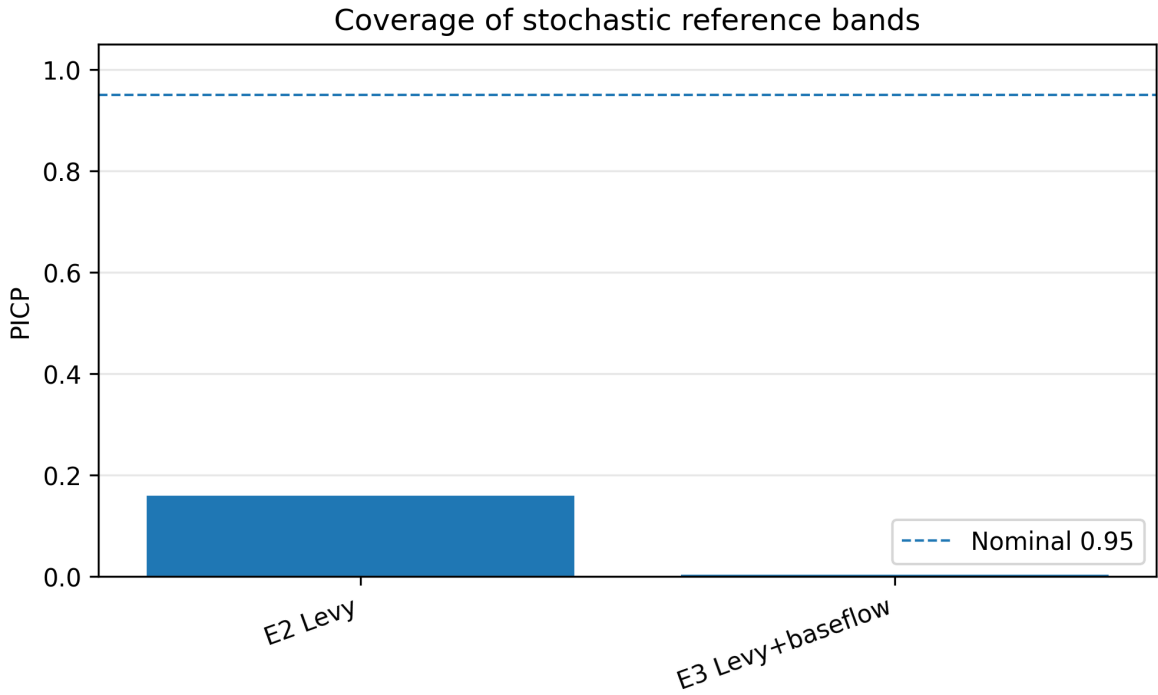


Figure 7: Prediction interval coverage probability for stochastic physical references. The dashed line marks the nominal 0.95 target.